

I. Induktion (2010)

$$1) \sum_{k=0}^n \left(\frac{1}{2}\right)^k = 2 - \frac{1}{2^n}$$

$$2) \sum_{k=0}^n q^k = \frac{q^{n+1} - 1}{q - 1} ; |q| < 1 ; n \geq 1$$

$$3) \sum_{k=0}^n (3k-2) = \frac{(1+n)(3n-4)}{2} ; n \geq 0$$

$$4) \sum_{k=2}^n (2k-4) = (n-2)(n-1)$$

$$5) 2^n > n^3 ; n \geq 10$$

$$6) n^2 \geq n+5 ; n \geq 3$$

$$7) 2n^3 + 3n^2 + n \text{ ist durch 6 teilbar}$$

$$8) 3^{2n} + 7 \text{ ist durch 8 teilbar}$$

$$9) \sum_{k=1}^n k \cdot 2^{k-1} > (n+1) \cdot n^2$$

$$10) \sum_{k=1}^n \frac{1}{k \cdot \sqrt{k}} < 3 - \frac{2}{\sqrt{n}} ; n \geq 2$$

$$11) \prod_{k=2}^{\infty} (1 - \frac{1}{k}) = \frac{1}{n}$$

$$12) \prod_{k=2}^{\infty} \left(1 + \frac{1}{k+1}\right) = 2 - \frac{1}{n+1}$$

- 1) $a_n = -\frac{3}{n} + 1$; $n \geq 1$
- 2) $a_n = \frac{3}{2n} + 4$; $n \geq 1$
- 3) $a_{n+1} = \sqrt[3]{3a_n + 5}$; $a_1 = 1$
- 4) $a_{n+1} = \sqrt[3]{2a_n + 25}$; $a_1 = 1$
- 5) $a_n = 3 + \left(\frac{2}{5}\right)^n$; $n \geq 0$
- 6) $a_n = \left(\frac{2}{3}\right)^n - 5$; $n \geq 0$
- 7) $a_{n+1} = \frac{1}{2} + 2a_n^2$; $a_1 = 0$
- 8) $a_{n+1} = \frac{2}{3} - 3a_n^3$; $a_1 = 0$
- 9) $a_n = \sqrt{\left(1 + \frac{4}{n}\right)^n} - \sqrt[n]{3^3}$; $n \geq 1$
- 10) $a_n = \sqrt[3]{\left(1 + \frac{6}{n}\right)^n} + \frac{2}{\sqrt[n]{2}}$; $n \geq 1$
- 11) $a_{n+1} = \sqrt{4a_n^3} + 1$; $a_1 = 1$; $n \geq 2$
- 12) $a_{n+1} = \frac{3}{\sqrt{2a_n + 1}}$; $a_1 = 0$

III Reihen (2010)

$$1) \sum_{k=3}^{\infty} \frac{4/3^k}{2} ; \sum_{k=5}^{10} 3 \cdot \left(\frac{2}{3}\right)^k$$

$$2) \sum_{k=2}^{\infty} (\sqrt{2} + e^{-2k}) ; \sum_{k=3}^5 \frac{1}{4} \cdot (2)^{-2k}$$

$$3) \sum_{k=1}^{\infty} (-2)^{k+1} \cdot \frac{\sqrt[k]{4}}{3^k}$$

$$4) \sum_{k=1}^{\infty} (-3)^{2k+1} \frac{2}{3^{4k}}$$

$$5) \sum_{k=1}^{\infty} \frac{\sqrt{k} \cdot (k+2)^2}{(3k)^k}$$

$$6) \sum_{k=1}^{\infty} 2^k \cdot \frac{(k+1)}{(2k-1)!}$$

$$7) \sum_{k=1}^{\infty} \frac{1 + \sin k}{(2+k)^k}$$

$$8) \sum_{k=1}^{\infty} \frac{k^2 \cdot 5^k}{(3k)!}$$

$$9) \sum_{k=3}^{\infty} \frac{2^{2k+1}}{3^k k!}$$

$$10) \sum_{k=2}^{\infty} (-1)^{k+1} \frac{2 \cdot 4^k}{(2k+1)!}$$

$$11) \sum_{k=3}^{\infty} \left(\left(\frac{1}{3}\right)^{2k} - \left(\frac{2}{1k}\right)^2 \right)$$

$$12) \sum_{k=3}^{\infty} \left(\frac{1}{k} \cdot \sqrt{3} \right)^2 - (2k)^{-2}$$

Wert
der
Reihe

IV Potenzreihen

(2010)

$$1) \sum_{k=1}^{\infty} \sqrt{4^{-k}} \cdot \left(\frac{4x}{3}\right)^k$$

$$2) \sum_{k=3}^{\infty} \sqrt[3]{\left(\frac{1}{8}\right)^k} \cdot (1,5x)^k$$

$$3) \sum_{k=0}^{\infty} \left(\frac{2k!}{(2k+1)!}\right)^3 \cdot x^k$$

$$4) \sum_{k=0}^{\infty} \frac{(k^2-1) \cdot x^k}{2^{2k+3}}$$

$$5) \sum_{k=1}^{\infty} \frac{81^k}{k^3} \cdot x^{4k}$$

$$6) \sum_{k=1}^{\infty} \frac{k^n \cdot x^k}{(2k)!} ; n \in \mathbb{N}$$

$$7) \sum_{k=1}^{\infty} \left(\frac{3}{7}x\right)^k = \sum_{k=0}^{\infty} \left(\frac{1}{3}\right)^{k+2} (2+x)^k$$

$$8) \sum_{k=1}^{\infty} \left(\frac{1}{4}x\right)^k = \sum_{k=0}^{\infty} \left(\frac{1}{5}\right)^{k+1} (x-1)^k$$

$$9) \sum_{k=1}^{\infty} \frac{(1-\sin x)^{2k}}{k^3}$$

$$10) \sum_{k=2}^{\infty} \frac{(\cos^2(x)+1)^{3k+1}}{(k!)^{4k}}$$

$$11) \sum_{k=0}^{\infty} 5 \cdot (2 \cdot x)^{3k} = 8$$

$$12) \sum_{k=2}^{\infty} \sqrt{\left(\frac{1}{4}\right)^k} = \int_1^{e^x} \frac{\cos(\ln x)}{x} dx$$

$$1) \lim_{x \rightarrow \pi/4} \frac{2\sin x - 2\cos x}{\cos 2x}$$

$$2) \lim_{x \rightarrow 0} \frac{\sin^2 x + 2\sin x}{x \cdot \cos x}$$

$$3) \lim_{x \rightarrow 0} \frac{1 - e^{2x}}{\sin x}$$

$$4) \lim_{x \rightarrow 0} \left(\frac{2}{x-2} - \frac{1}{e^x - 1} \right)$$

$$5) \lim_{x \rightarrow -1} \frac{x^3 + 2x^2 - x + 2}{3x^3 + 3x^2 - 3x - 3}$$

$$6) \lim_{x \rightarrow 3} \frac{x^3 + 2x^2 - 9x - 18}{2x^3 - 2x^2 - 18x + 18}$$

$$7) \lim_{x \rightarrow 0} \frac{2x}{\sqrt{4+x} - 2}$$

$$8) \lim_{x \rightarrow 0} \frac{-3(x-1)}{\sqrt{3+6x} - 3}$$

$$9) \lim_{x \rightarrow 0} \frac{\ln(1 + \ln(x+1))}{\sin x}$$

$$10) \lim_{x \rightarrow \pi} \frac{\cos(x/2 + \pi)}{\tan x}$$

VI. Funktionen

(2010)

1) $f(x) = \frac{3}{2} \cdot \cos(2x + 4\pi) - 1$

2) $f(x) = -4 \cdot \sin(5x - \frac{3}{2}\pi) + 2$

3) $f(x) = -2 \cdot (\cos^2(x + 3\pi))^2 - 4$

4) $f(x) = 3 \cdot (\sin^3(\frac{1}{3}x - \frac{\pi}{2}))^4 + 2$

- Symmetrie
- Amplituden (H_f)
- Periode

5) $f(x) = \frac{1}{5} \cdot \ln(x^3 - 8) - 2$

6) $f(x) = -2 \cdot e^{(x-2)^2} + 3$

- ① und IV
- Umkehrfunktion $f^{-1}(x)$
- Monotonie

7) $f(x) = \frac{3x^2 \cdot \cosh x}{1 - \sqrt{1+x^2}}$

8) $f(x) = 2 \cdot |x - 4|$

- stetig
- differenzierbar

} Df

9) $f(x) = \frac{2x-6}{x^2-5x+6} + 1$

10) $f(x) = \frac{6-2x}{x^2-7x+12} - 1$

- Definitionsbereich
- Asymptoten
- Skizze

11) $f(x) = \arctan(x)$

12) $f(x) = \cot(2x)$

- Definitionsbereich
- Wertebereich
- Symmetrie

VIII Ableitungen

(2010)

$$1) f(x) = \arctan(e^{\sqrt{x^4+1}} - x^2)$$

$$2) f(x) = -1/3 \cdot \sinh[\ln(x^{-2} + x)]^2$$

$$3) f(x) = 3^{\ln x} \cdot \sin(2x)$$

$$4) f(x) = 2^{\cos(2x)} \cdot \ln x^2$$

$$5) f(x) = \frac{\sin^2(x) + 3x}{\ln(2x - x^3)}$$

$$6) f(x) = \frac{e^{4-x^2}}{\tan(3-x)}$$

$$7) f(x) = x^2 \cdot e^{-x}$$

$$8) f(x) = x^3 \cdot \ln(1/x^2)$$

} $\rightarrow$ globale Extremstellen für $x \geq 0$

$$9) f(x) = \sin(\ln x^3) \quad ; \quad f'(x_0) = ? \quad ; \quad x_0 = \sqrt[3]{2}$$

$$10) f(x) = (2x)^{\cos(\sqrt{x})} \quad ; \quad f'(x_0) = ? \quad x_0 = \pi/4$$

VIII Satz von Taylor (2010)

1) $f(x) = \sqrt[4]{2x+1}$; $P_3(x, 0)$

2) $f(x) = \frac{3}{\sqrt[3]{2x-1}}$; $P_3(x, 0)$

3) $f(x) = 2x^3 - 2 \cdot e^{2x}$; $P_3(x, 0)$

4) $f(x) = \ln \sqrt{2x-1} - \frac{1}{3}x^2$; $P_3(x, 1)$

5) $f(x) = \sin(2\pi x) - 2x$; $P_3(x, \frac{1}{2})$

6) $f(x) = 4 \cdot \cos(\frac{\pi}{4} \cdot x) + 2 \cdot x^3$; $P_3(x, 0)$

- Taylor - Polynom
- Fehlerabschätzung / Restglied
- Definitionsbereich - / Wertebereich
- Umkehrfunktion $f^{-1}(x)$
- Ableitung $(f^{-1}(x))'$

7) $f(x) = \sqrt{x^{-3}}$; $P_4(x, 1)$

8) $f(x) = 2 \cdot \sqrt[3]{x^2}$; $P_4(x, 1)$

IX. Integralrechnung

(2010)

1) $f(x) = x^2 - x - 12$; $g(x) = 15 - 2x$

2) $f(x) = x^2 + 5x - 6$; $g(x) = 2x + 4$

3) $\int_{-6}^0 x^3 + 6x^2 - x - 30 \, dx$

4) $\int_0^5 x^3 - 19x + 30 \, dx$

5) $\int_{-\pi}^{\pi} \frac{x \cdot e^{-x^3}}{x^{-1}} \, dx$

6) $\int_{-1}^2 \frac{2 \, dx}{x^2 + 16} \, dx$

7) $\int \sin(2x) \cdot \cosh(1/4 x) \, dx$

8) $\int \frac{1 - \cos(x)}{e^{3x}} \, dx$

9) $\int 1/3 x^2 \cdot \sinh(6x) \, dx$

10) $\int 2x^3 \cdot e^{-2x} \, dx$

11) $\int_1^2 \frac{e^{2x}}{4 - 3e^{2x}} \, dx$

12) $\int_{-e}^e x \cdot \sqrt{x^2 + 1} \, dx$